Drosophila Dual-Goal Steering Madness

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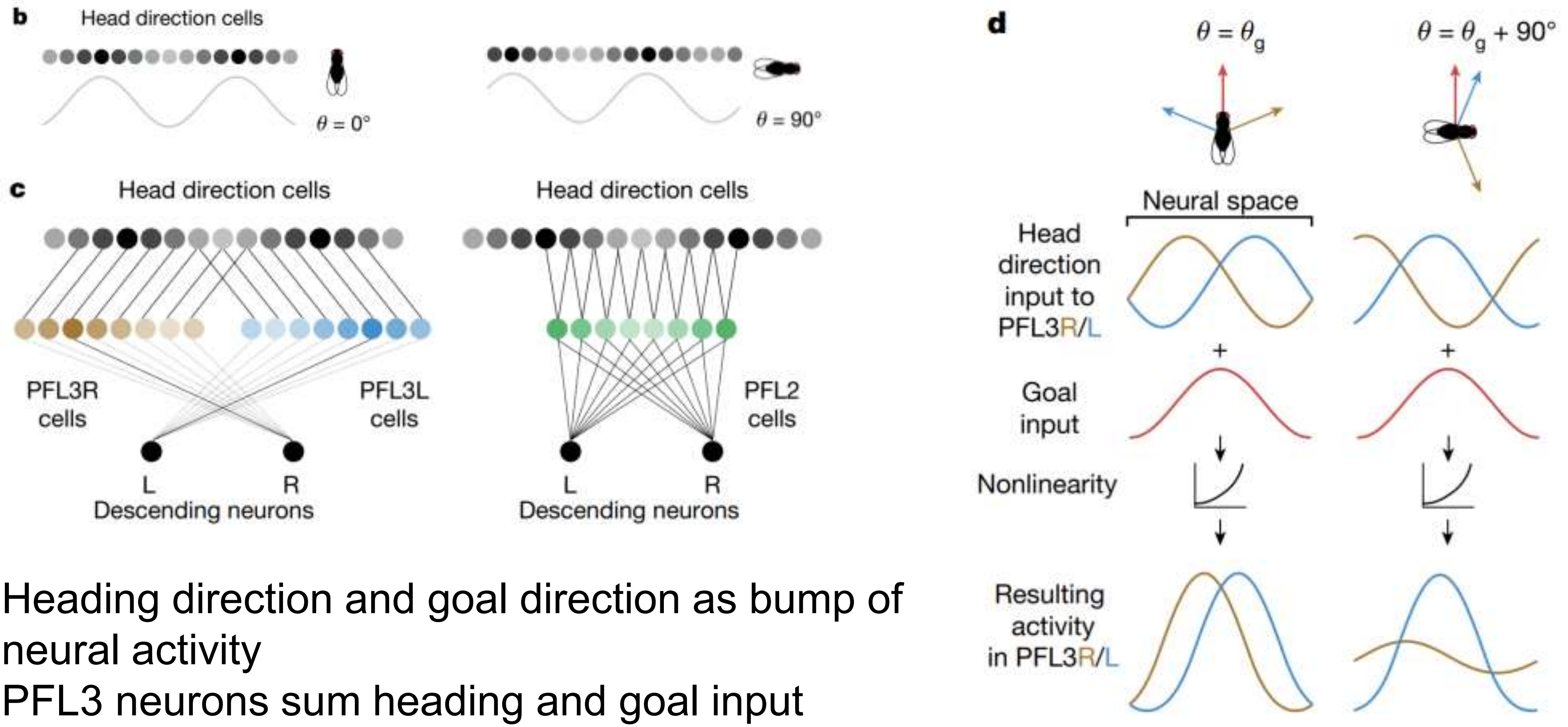
The Central Complex Steering Circuit



- Heading direction and goal direction as bump of neural activity
- PFL3 neurons sum heading and goal input
- Different between sum of left and right PFL activity determines steering command

Westeinde, et al., 2024

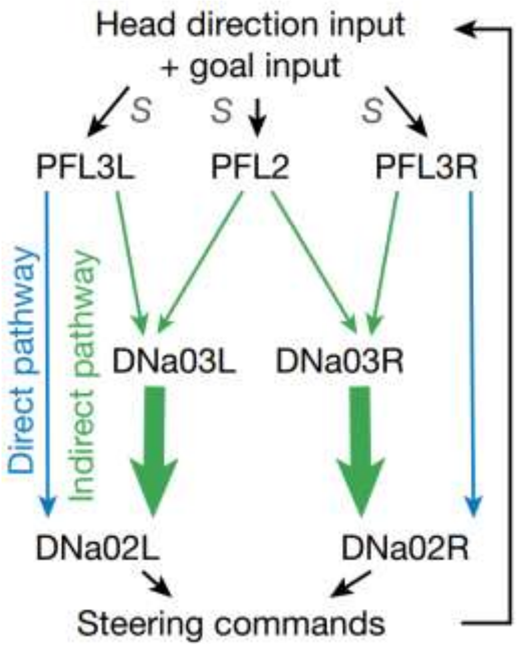
The Full Circuit Model

- Heading direction and goal direction as bump of neural activity
- PFL3 neurons sum heading and goal input

$$\mathrm{PFL2} = f(S \times (\cos(\theta - \theta_0 - \mathbf{h} + 180°) + A \times \cos(\theta_g - \theta_0 - \mathbf{h})))$$
$$\mathrm{PFL3R} = f(S \times (\cos(\theta - \theta_0 - \mathbf{h} + 67.5°) + A \times \cos(\theta_g - \theta_0 - \mathbf{h})))$$
$$\mathrm{PFL3L} = f(S \times (\cos(\theta - \theta_0 - \mathbf{h} - 67.5°) + A \times \cos(\theta_g - \theta_0 - \mathbf{h})))$$

$$\mathrm{ELU}(M) = M \text{ for } M \geq 0$$

$$\mathrm{ELU}(M) = \mathrm{e}^M - 1 \text{ for } M < 0$$



- Different between sum of left and right PFL activity determines steering command

$$
\begin{aligned}
&d\theta/dt \propto \mathrm{DNa02R} - \mathrm{DNa02L} + \epsilon \\
&= f(\Sigma W_{\mathrm{D2R,P3R}_j} \times \mathrm{P3R}_j + \Sigma W_{\mathrm{D2,D3}} \times \mathrm{D3R}) \\
&\quad - f(\Sigma W_{\mathrm{D2L,P3L}_j} \times \mathrm{P3L}_j + \Sigma W_{\mathrm{D2,D3}} \times \mathrm{D3L}) + \epsilon
\end{aligned}
$$

Westeinde, et al., 2024

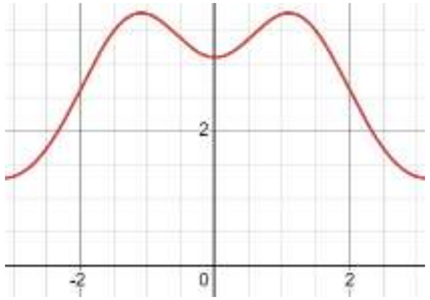
Modifications for two goals

- Heading direction and goal direction as bump of neural activity

$$g_{\text{heading}}(\phi,\theta) = \exp(\kappa_{\text{heading}}\cos(\theta-\phi)) \qquad g_{\text{goal}}(\phi,\delta) = \exp(\kappa_{\text{goal}}\cos(\delta-\phi)) + \exp(\kappa_{\text{goal}}\cos(-\delta-\phi))$$

- PFL3 neurons sum heading and goal input

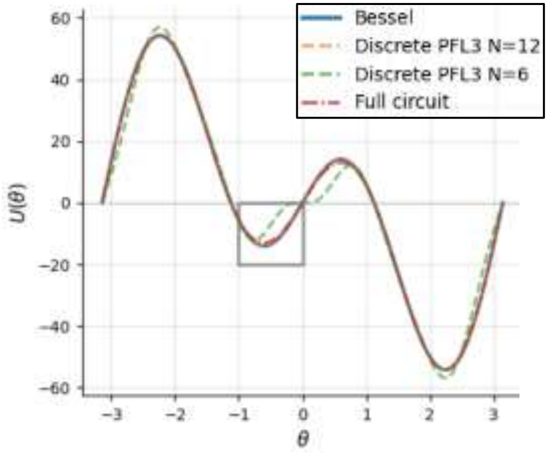
$$\begin{aligned} h_{PFL3L}(\theta,\phi,\delta) &= S\cdot g_{\text{heading}}(\phi-\Delta_{PFL3},\theta) + A\cdot g_{\text{goal}}(\phi,\delta) \\ &= S\cdot\exp(\kappa_{\text{heading}}\cos(\theta-(\phi-\Delta_{PFL3}))) \\ &\quad +A\cdot\underline{(\exp(\kappa_{\text{goal}}\cos(\delta-\phi))+\exp(\kappa_{\text{goal}}\cos(-\delta-\phi)))} \end{aligned}$$



- Different between sum of left and right PFL activity determines steering

$$\begin{aligned} G_{PFL3L}(\theta,\delta) &= \int_{-\pi}^{\pi} h_{PFL3}(\theta,\phi,\delta)^2\,d\phi \\ &= \int_{-\pi}^{\pi} (S\cdot g_{\text{heading}}(\phi-\Delta_{PFL3},\theta)+A\cdot g_{\text{goal}}(\phi,\delta))^2\,d\phi \\ &= \int_{-\pi}^{\pi} \big[S^2\cdot g_{\text{heading}}(\phi-\Delta_{PFL3},\theta)^2 + A^2\cdot g_{\text{goal}}(\phi,\delta)^2 \\ &\quad +2SA\cdot g_{\text{heading}}(\phi-\Delta_{PFL3},\theta)\cdot g_{\text{goal}}(\phi,\delta)\big]\,d\phi \end{aligned}$$

Derivation of the Reduced System



$$
\begin{aligned}
G_{PFL3L}(\theta,\delta) &= \int_{-\pi}^{\pi} h_{PFL3}(\theta,\phi,\delta)^2\, d\phi \\
&= \int_{-\pi}^{\pi} \left(S \cdot g_{\text{heading}}(\phi - \Delta_{PFL3}, \theta) + A \cdot g_{\text{goal}}(\phi,\delta)\right)^2 d\phi \\
&= \int_{-\pi}^{\pi} \Big[S^2 \cdot g_{\text{heading}}(\phi - \Delta_{PFL3}, \theta)^2 + A^2 \cdot g_{\text{goal}}(\phi,\delta)^2 \\
&\quad + 2SA \cdot g_{\text{heading}}(\phi - \Delta_{PFL3}, \theta) \cdot g_{\text{goal}}(\phi,\delta)\Big]\, d\phi
\end{aligned}
$$

Let $\xi_\phi(a,b) : S^1 \times S^1 \to \mathbb{R} = \exp(\kappa_{\text{heading}} \cos(a-\phi) + \kappa_{\text{goal}} \cos(b-\phi))$, we can rewrite the above as:

$$
\begin{aligned}
G_{PFL3L}(\theta,\delta) - G_{PFL3R}(\theta,\delta) = 2SA \int_{-\pi}^{\pi} &\big(\xi_\phi(\theta + \Delta_{PFL3}, \delta) + \xi_\phi(\theta + \Delta_{PFL3}, -\delta) \\
&- \xi_\phi(\theta - \Delta_{PFL3}, \delta) - \xi_\phi(\theta - \Delta_{PFL3}, -\delta)\big) d\phi
\end{aligned} \tag{7}
$$

Thanks to the phasor sum & Bessel function

$$
\begin{aligned}
G_{PFL3L}(\theta,\delta) - G_{PFL3R}(\theta,\delta) = 4\pi SA\big(&I_0(\kappa(\theta + \Delta_{PFL3}, \delta)) + I_0(\kappa(\theta + \Delta_{PFL3}, -\delta)) \\
&- I_0(\kappa(\theta - \Delta_{PFL3}, \delta)) - I_0(\kappa(\theta - \Delta_{PFL3}, -\delta))\big)
\end{aligned}
$$

Taylor Expansion Approximation

$$
\begin{aligned}
G_{PFL3L}(\theta,\delta) - G_{PFL3R}(\theta,\delta) = 4\pi SA\big(&I_0(\kappa(\theta+\Delta_{PFL3},\delta)) + I_0(\kappa(\theta+\Delta_{PFL3},-\delta)) \\
&-I_0(\kappa(\theta-\Delta_{PFL3},\delta)) - I_0(\kappa(\theta-\Delta_{PFL3},-\delta))\big)
\end{aligned}
$$

$$U(\theta,\delta) \approx U'(0,\delta)\,\theta + \frac{1}{6}U'''(0,\delta)\,\theta^3 + O(\theta^5).$$

Thanks to symmetric property of kappa

$$\boxed{U'(0,\delta) = -8\pi SA\,\kappa_h\kappa_g\left[\frac{I_1(\kappa(\Delta_{PFL3},\delta))}{\kappa(\Delta_{PFL3},\delta)}\sin(\Delta_{PFL3}-\delta) + \frac{I_1(\kappa(\Delta_{PFL3},-\delta))}{\kappa(\Delta_{PFL3},-\delta)}\sin(\Delta_{PFL3}+\delta)\right]}$$

$$\boxed{U'''(0,\delta) = 8\pi SA\kappa_h\kappa_g[\sin(\Delta_{PFL3}-\delta)\cdot D(\kappa_-,C_-,S_-) + \sin(\Delta_{P_3L}+\delta)\cdot D(\kappa_+,C_+,S_+)]}$$

with

$$D(K,C,S) = \frac{1}{K}\left[I_1(K)\left(1-\frac{3C}{K^2}-\frac{3S^2}{K^4}\right) + \frac{3(I_0(K)+I_2(K))}{2}\left(\frac{C}{K}+\frac{S^2}{K^3}\right) - \frac{3I_1(K)+I_3(K)}{4}\cdot \ldots\right.$$

and

$$
\begin{aligned}
&\kappa_- \equiv \kappa(\Delta_{PFL3},\delta) = \sqrt{\kappa_h^2+\kappa_g^2+2\kappa_h\kappa_g\cos(\Delta_{PFL3}-\delta)} \\
&\kappa_+ \equiv \kappa(\Delta_{PFL3},-\delta) = \sqrt{\kappa_h^2+\kappa_g^2+2\kappa_h\kappa_g\cos(\Delta_{PFL3}+\delta)} \\
&C_- \equiv \kappa_h\kappa_g\cos(\Delta_{PFL3}-\delta), \qquad C_+ \equiv \kappa_h\kappa_g\cos(\Delta_{PFL3}+\delta) \\
&S_- \equiv \kappa_h\kappa_g\sin(\Delta_{PFL3}-\delta), \qquad S_+ \equiv \kappa_h\kappa_g\sin(\Delta_{PFL3}+\delta)
\end{aligned}
$$

Body Mechanics and Second-Order System

$\dot{\theta} = U(\theta,\delta) \approx A(\delta)\,\theta - B(\delta)\,\theta^3 + O(\theta^5)$, with

$$A(\delta) \equiv U'(0,\delta), \qquad B(\delta) \equiv -\frac{1}{6}U'''(0,\delta).$$

$$\boxed{U'(0,\delta) = -8\pi S A\,\kappa_h\kappa_g\left[\frac{I_1(\kappa(\Delta_{PFL3},\delta))}{\kappa(\Delta_{PFL3},\delta)}\sin(\Delta_{PFL3}-\delta) + \frac{I_1(\kappa(\Delta_{PFL3},-\delta))}{\kappa(\Delta_{PFL3},-\delta)}\sin(\Delta_{PFL3}+\delta)\right]}$$

$$\boxed{U'''(0,\delta) = 8\pi S A\kappa_h\kappa_g[\sin(\Delta_{PFL3}-\delta)\cdot D(\kappa_-,C_-,S_-) + \sin(\Delta_{P_3L}+\delta)\cdot D(\kappa_+,C_+,S_+)]}$$

Angular velocity command

Overdamping $I \to 0$

Torque command

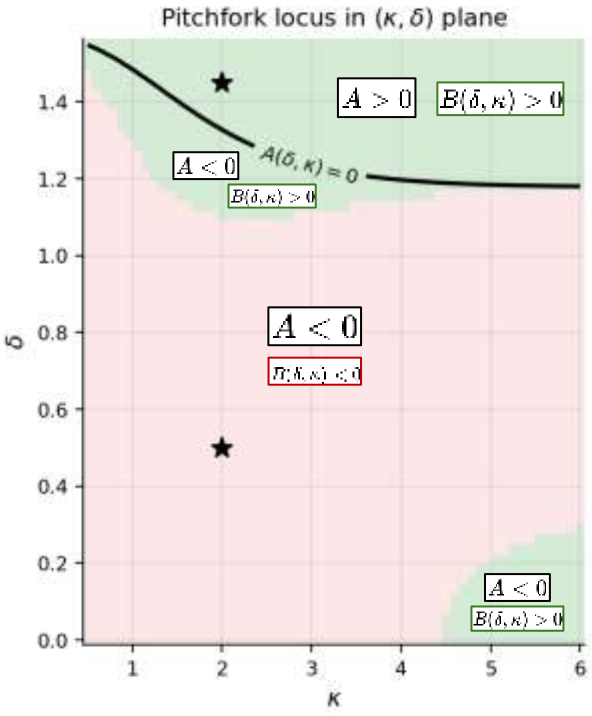
$$I\ddot{\theta} + c\dot{\theta} = A(\delta,\kappa)\,\theta - B(\delta,\kappa)\,\theta^3$$

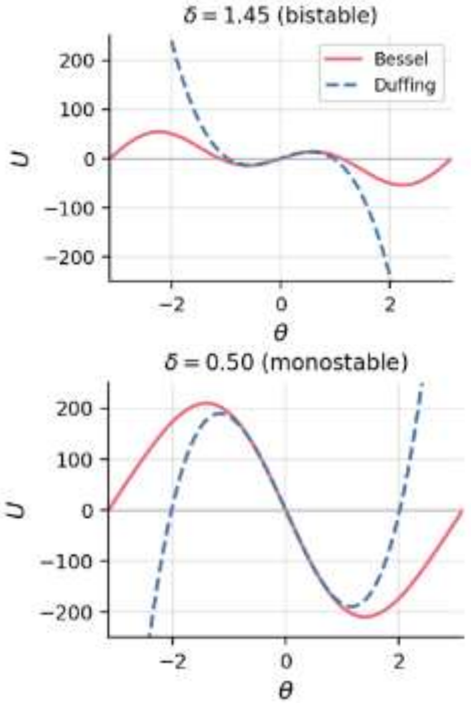
Double-well Energy Landscape

$$I\ddot{\theta} + \gamma\dot{\theta} = A(\delta,\kappa)\,\theta - B(\delta,\kappa)\,\theta^3$$

Linearize at the origin, $\dot{\theta} = A\theta$, so the fixed point at $\theta = 0$ is **unstable** iff. $A(\delta,\kappa) = U'(0,\delta) > 0$

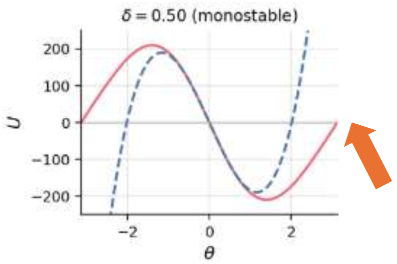
When in addition $B(\delta,\kappa) > 0$ (i.e. $U'''(0,\delta) < 0$), two new equilibria emerge at $\theta = \pm\sqrt{\frac{A}{B}}$

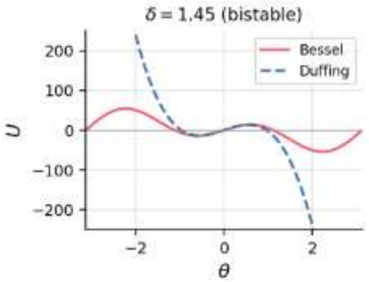




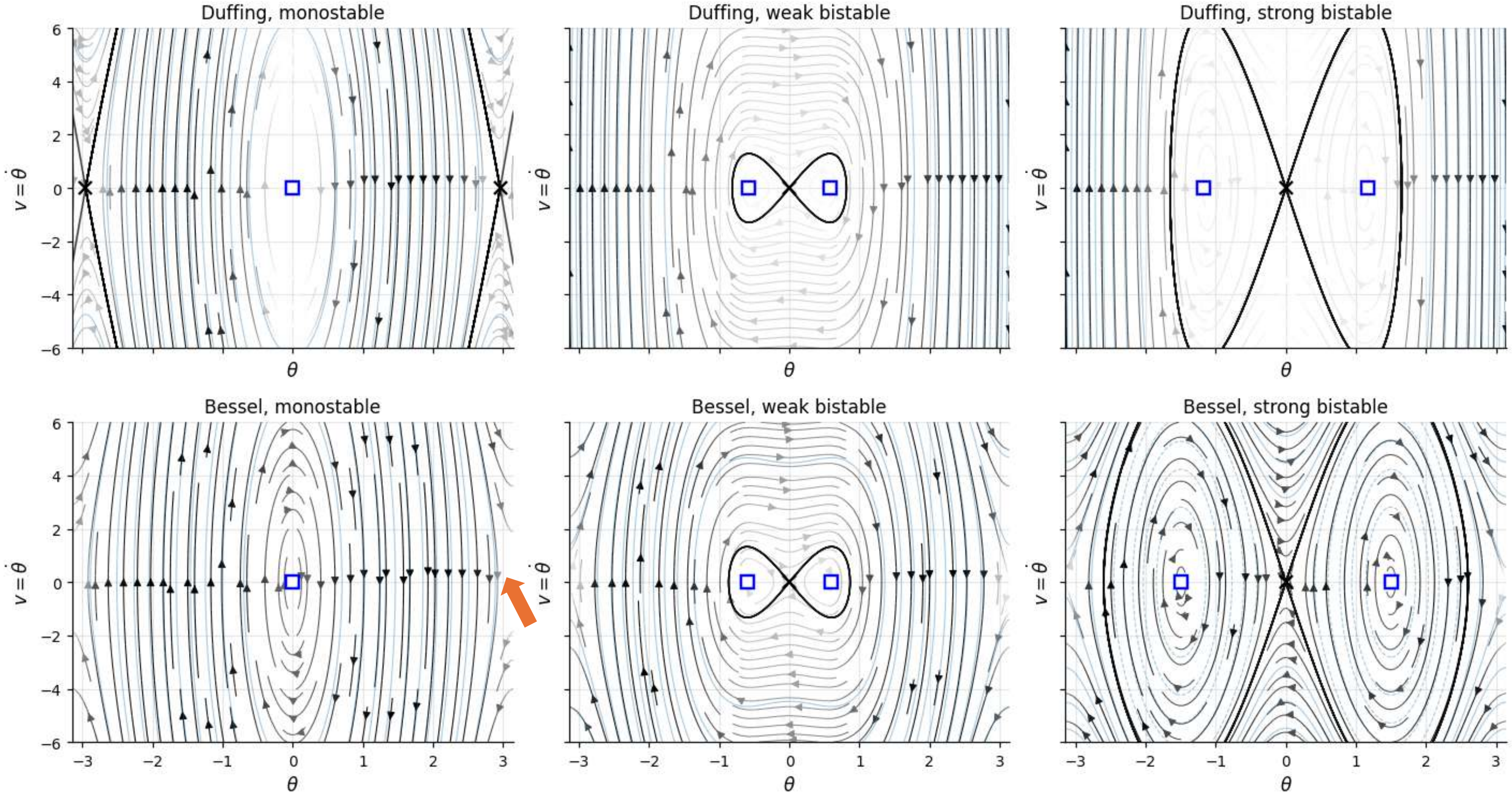
Kappa = 2

Delta = 1, 1.36, 1.55





Phase Portrait



$\boxed{A < 0}$ $\boxed{B(\delta,\kappa) < 0}$

$\boxed{A > 0}$ $\boxed{B(\delta,\kappa) > 0}$

Homoclinic Orbits

$$H(\theta, v) = \frac{1}{2}v^2 - \frac{1}{2}A\theta^2 + \frac{1}{4}B\theta^4, \qquad v \equiv \dot{\theta}.$$

$$= 0$$

$$\downarrow$$

$$\boxed{\theta_0(t) = \sqrt{\frac{2A}{B}}\,\mathrm{sech}(\sqrt{A}\,t), \qquad v_0(t) = -\sqrt{\frac{2A}{B}}\sqrt{A}\,\mathrm{sech}(\sqrt{A}\,t)\tanh(\sqrt{A}\,t).}$$

Melnikov's Method

$$\dot{\theta} = v, \qquad \dot{v} = A\theta - B\theta^3 + \epsilon\left[-\gamma v + F\cos\omega t\right],$$

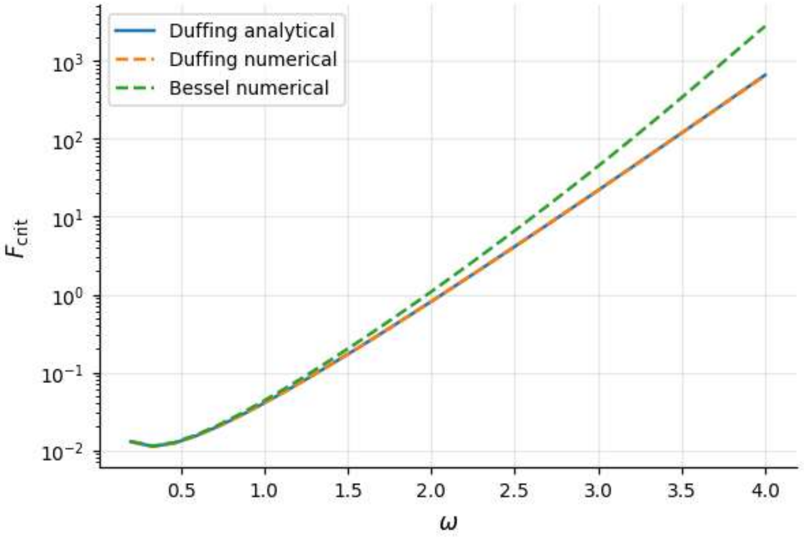
$$\mathbf{g} = (0, -\gamma v + F\cos\omega t)^T$$

$$\boldsymbol{\gamma}_0 = (\theta_0, v_0)^T$$

$$H(\theta, v) = \frac{1}{2}v^2 - \frac{1}{2}A\theta^2 + \frac{1}{4}B\theta^4, \qquad v \equiv \dot{\theta}.$$

$$\begin{aligned} M(t_0) &= \int_{-\infty}^{\infty} \nabla H(\boldsymbol{\gamma}_0(\tau)) \cdot \mathbf{g}(\boldsymbol{\gamma}_0(\tau), \tau + t_0)\, d\tau \\ &= -\frac{4\gamma A^{3/2}}{3B} - F\pi\omega\sqrt{\frac{2}{B}}\,\operatorname{sech}\left(\frac{\pi\omega}{2\sqrt{A}}\right)\sin\omega t_0 \\ &= 0 \end{aligned}$$

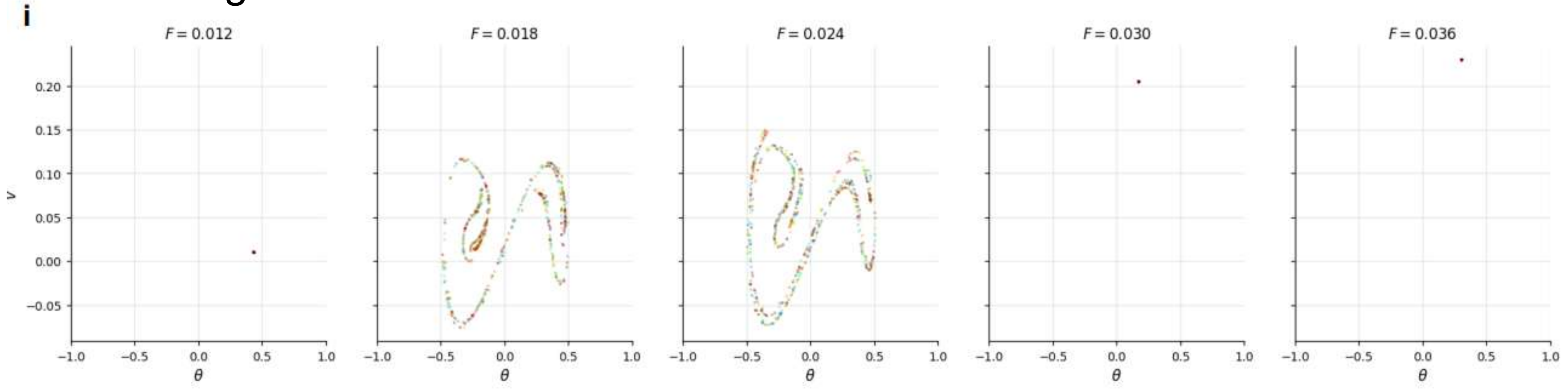
$$\boxed{F_{\text{crit}}(\omega) = \frac{4\gamma A}{3\pi\omega\sqrt{2B}}\cosh\left(\frac{\pi\omega}{2\sqrt{A}}\right).}$$

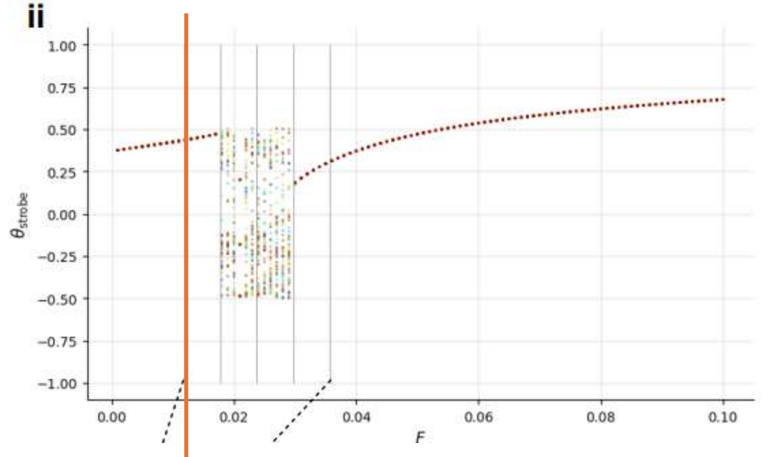


Numerical Validations – Vary Forcing Amplitude

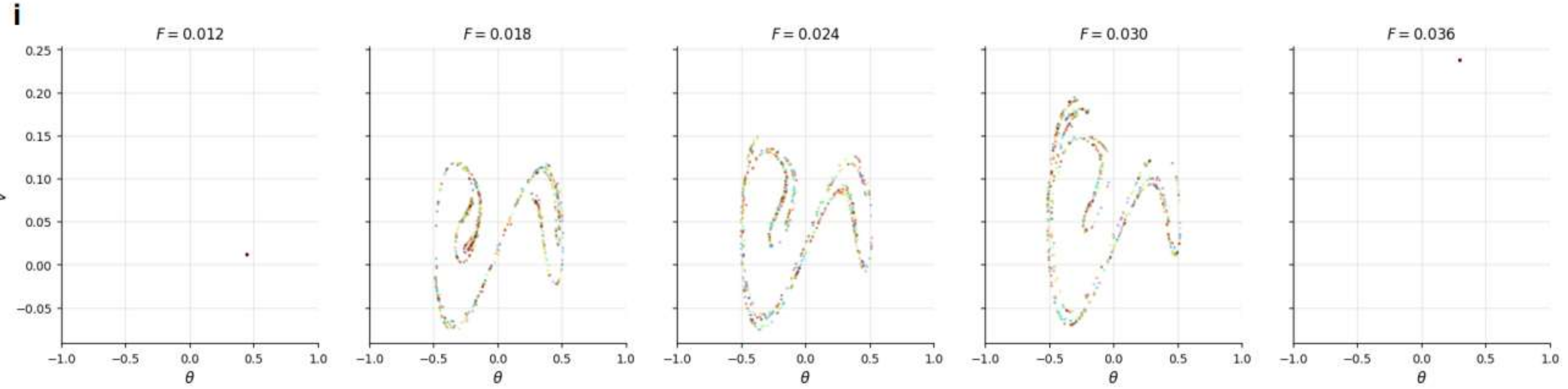
Duffing

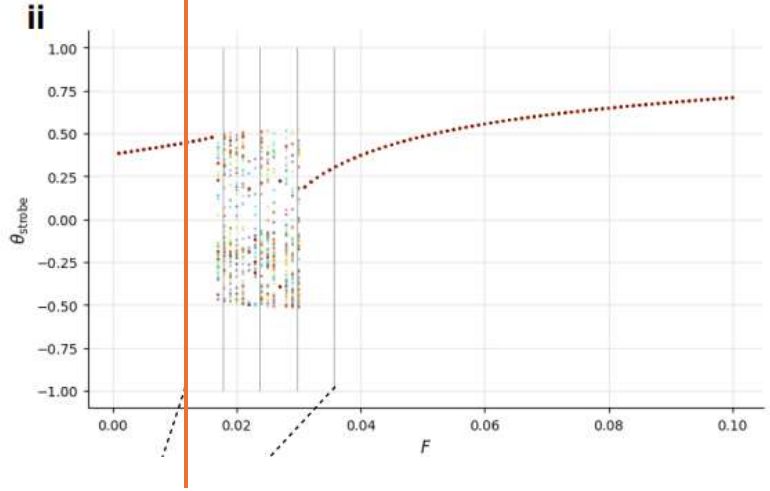
$F_{critical}$



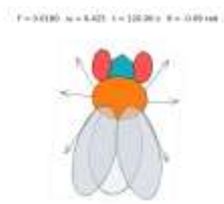


12 PFL3 Neurons



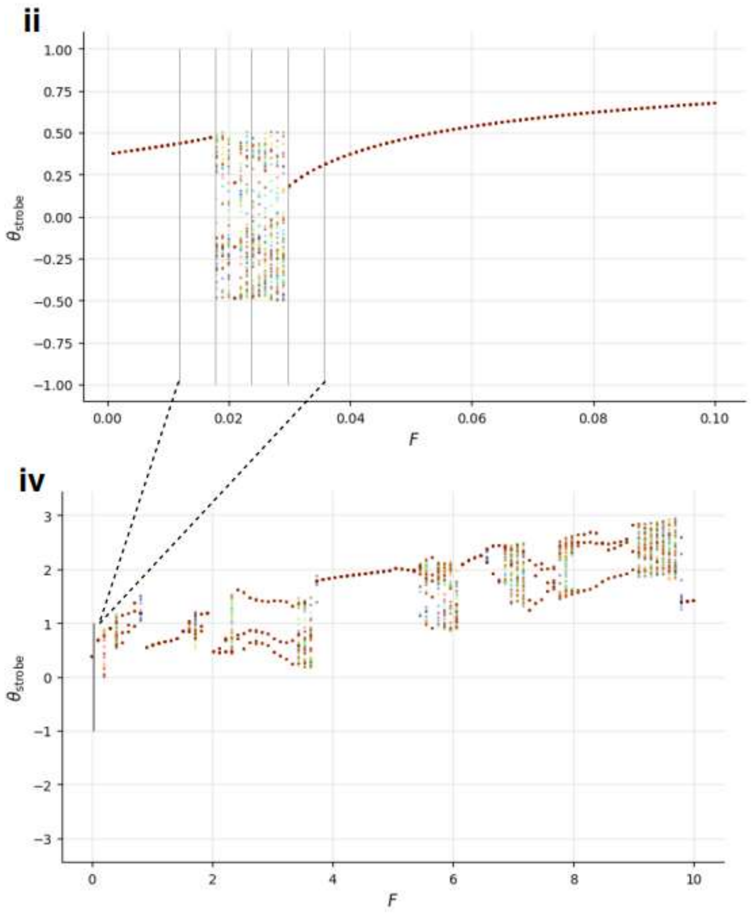


Chaos

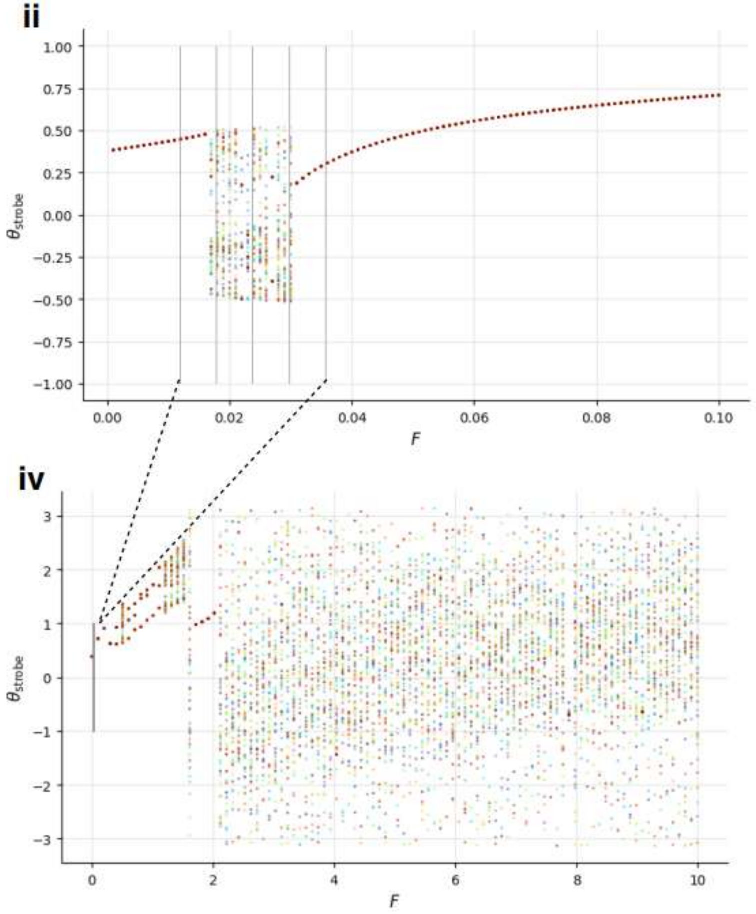


Numerical Validations

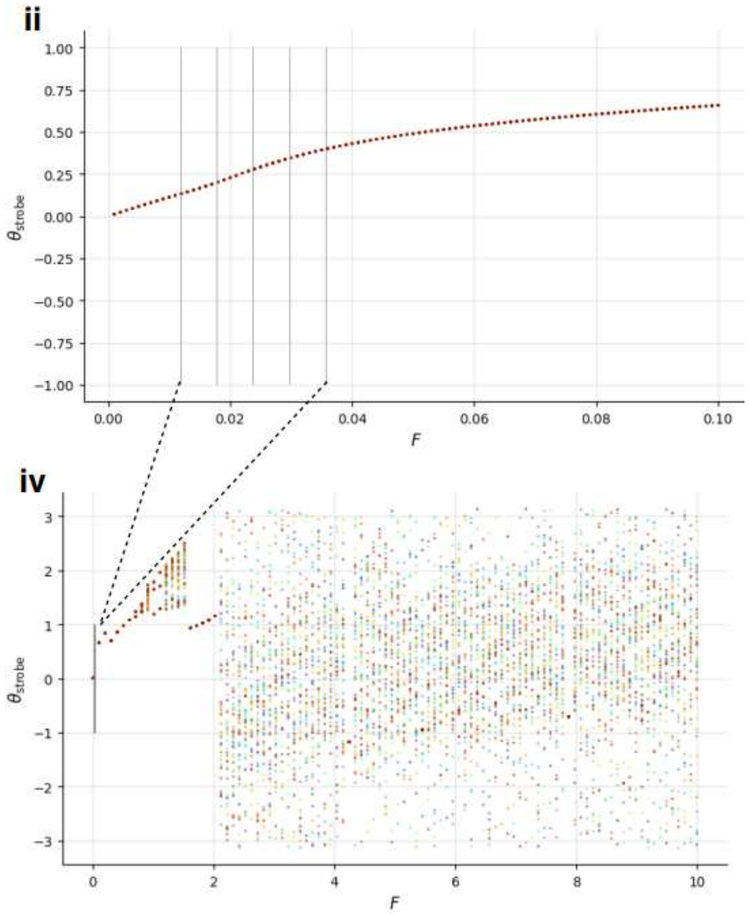
Duffing



12 PFL3 Neurons



6 PFL3 Neurons



Chaos

Numerical Validations

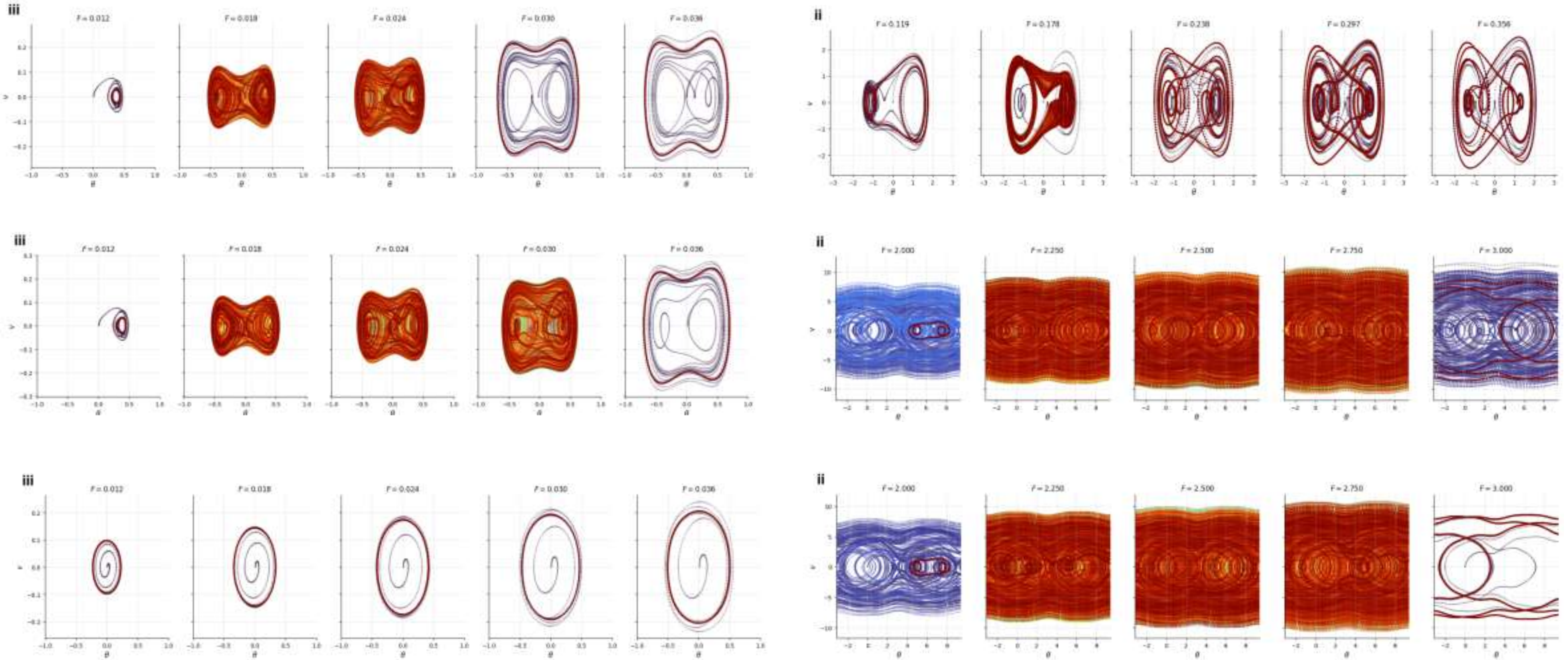
F = 0.012 - 0.036

F = 2 - 3

Duffing

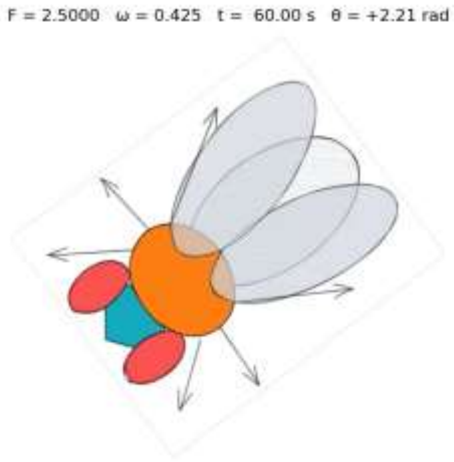
12 PFL3 Neurons

6 PFL3 Neurons



Summary

- The neural circuit model under dual goal condition has a double-well potential
- We connected it to the Duffing equation using Taylor expansion
- We analyzed the bifurcation conditions
- We derived the homoclinic orbit and solved for the conditions for chaos
- We made cartoons when the fly goes nuts



Thank you!